(1) Binary Cross Entropy

$$L = -[y \cdot \log(a) + (1-y) \cdot \log(1-a)]$$

a = prediction , y = target

$$\frac{dL}{da} = \frac{d}{da}\left[-y\log(a) - (1-y)\log(1-a)\right]$$

$$= -y\frac{1}{a} - (1-y)\frac{1}{1-a}$$

$$= \frac{-y}{a} + \frac{(1-y)}{1-a}$$

$$= \frac{-y + ay + a - ay}{a(1-a)}$$

$$= \frac{a-y}{a(1-a)}$$